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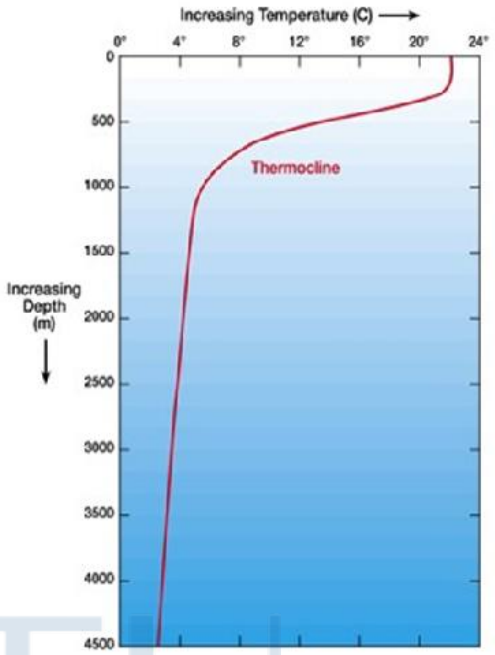
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Module 4

Ocean Thermal Energy Conversion

Ocean Thermal Energy Conversion (OTEC) is a process which utilizes the heat energy stored in the tropical ocean. OTEC Process uses temperature difference between cold deep water (5^0C) & warm surface water (27^0C)to power a turbine to generate electricity.



It uses the ocean's naturally available temperature gradient (thermocline), or thermal energy. Because the oceans are continually heated by the sun and cover nearly 70% of the Earth's surface, this temperature difference contains a vast amount of solar energy which could potentially be tapped for human use.

Oceans collect 80% of the incident solar energy. Up to 88,000 TWh/yr of power could be generated from OTEC without affecting the ocean's thermal structure. Unlike wind or solar energy, ocean thermal energy is available throughout the day, thus eliminating the need to store energy.

Basic Principle

In the ocean maximum temperature exists just below the upper surface due to solar heating and the lowest temperature exists at deep water due to the absence of solar heating. Warm water at the top is a heat source at a temperature of about 27^0 C and the water at the bottom is a heat sink at about 4^0 C. A heat engine can work between this heat source and heat sink to convert heat into mechanical work and then into electrical energy.

If Source temperature (T_1) is the temperature of warm surface water of ocean and Sink temperature (T_2) is the temperature of cold deep water of ocean. Maximum theoretical efficiency of a heat engine operating between source temperature T_1 and sink temperature T_2 is called Carnot's efficiency.

$$\Psi = \frac{T1 - T2}{T1}$$

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Classification



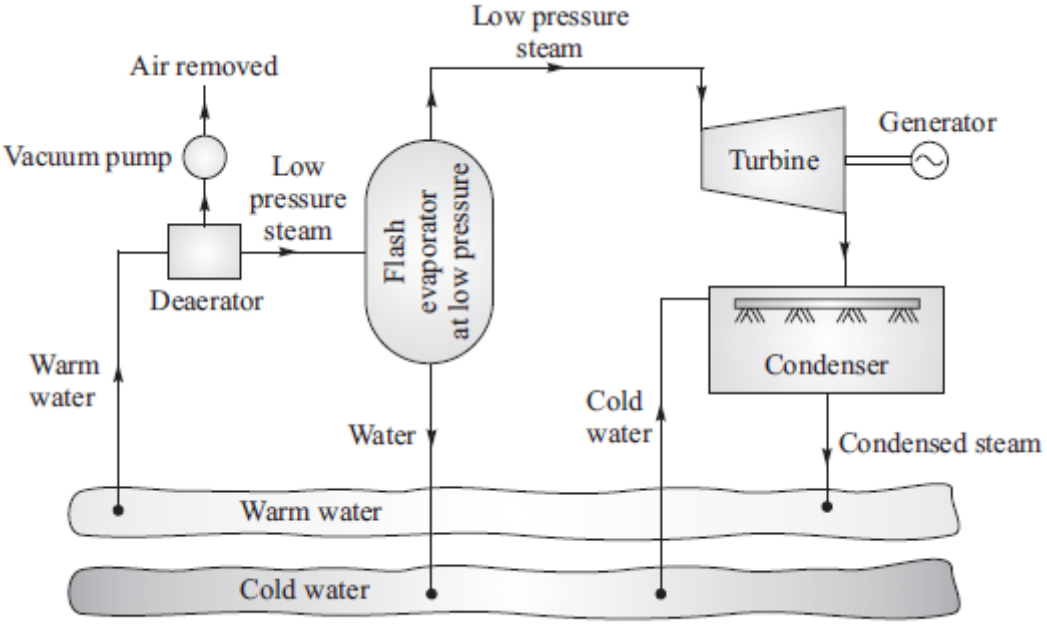
Open Cycle OTEC (Claude System)

Open cycle refers to the utilization of sea water as the working fluid where in sea water is flash evaporated under a partial vacuum. The warm surface water is caused to boil by lowering the pressure, without supplying any additional heat. The low pressure steam is passed through a turbine which extracts energy from it and then the vapour is cooled in a condenser.

A heat exchanger is not required in the evaporator & direct contact between the exhaust steam & a cold water spray makes a heat exchanger necessary in the condenser. This cycle is called open cycle as the condensate need not be returned to the evaporator as in case of the closed cycle. Instead, the condensate can be utilized as desalinated water if a surface condenser is used.

In this cycle, warm sea water at 27^0C is admitted into an evaporator in which the pressure is maintained at a value slightly below the saturation pressure corresponding to that temperature. Because of the need in the open cycle to harness energy in low pressure steam, extremely large turbines must be utilized. Furthermore, degasifiers (deaerators) must be used to remove the gases dissolved in the sea water. On the other hand, since there are no heat transfer problems in the evaporator, the problem of bio–fouling control is minimized.

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Open cycle OTEC plant.

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Closed Cycle OTEC (Anderson Cycle)

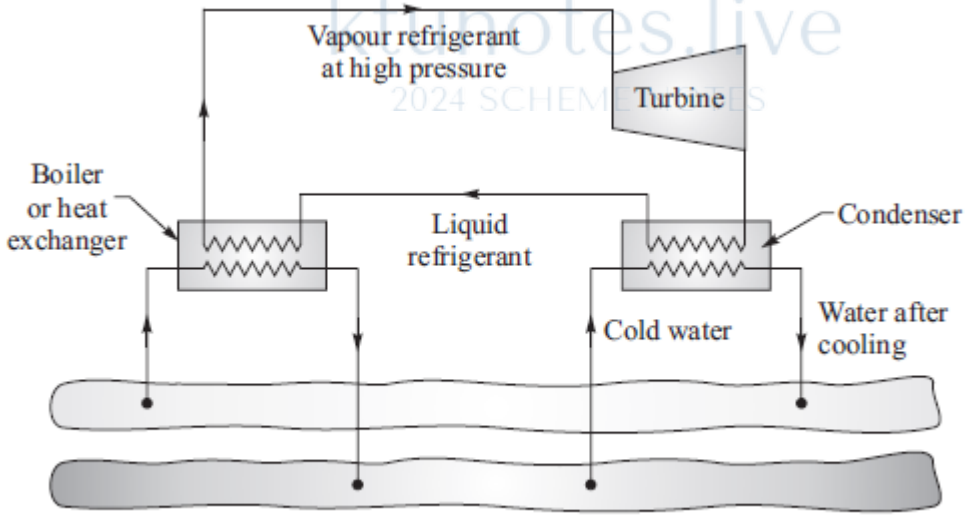
In the closed cycle system, a liquid working fluid such as ammonia or propane is vaporized in an evaporator or boiler. The heat required for vaporization is transferred from the warm ocean surface to the liquid by means of a heat exchanger. Heat exchangers such as evaporators and condensers are a key ingredient since large volume of water must be circulated through the OTEC power plant.

This cycle requires a separate working fluid (other than sea water) that receives and rejects heat to the source and sink via heat exchangers. A fundamental requirement in closed cycle system is to transfer heat efficiently across the heat exchanger surfaces so as to achieve a high value of overall heat transfer coefficient.

The high pressure leaving the evaporator drives an expansion turbine that is designed to operate at a lower inlet pressure. The turbine is connected to an electric generator to convert the mechanical energy produced to electrical energy. The low pressure exhaust from the turbine is cooled and converted back into liquid in the condenser.

The cooling is achieved by passing cold, deep ocean water from a depth of 700-900m or more through a heat exchanger. The liquid working fluid is then pumped back as high pressure liquid to the evaporator thus closing the cycle.

To maintain viable OTEC heat exchangers, provisions must be made to inhibit the formation of fouling layers and to remove any significant fouling that forms. The removal can be accomplished by periodically cleaning the heat exchanger surfaces through mechanical, chemical or other means

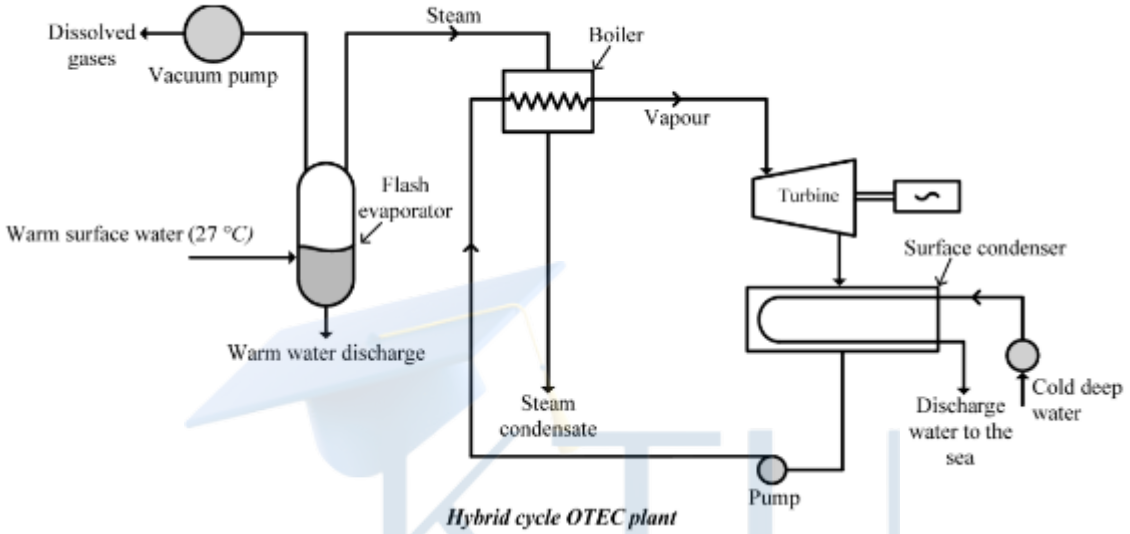


Closed cycle OTEC plant.

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Hybrid Cycle OTEC

A hybrid cycle combines the features of both the closed-cycle and open-cycle systems. Warm seawater enters a vacuum chamber where it is flash-evaporated into steam, which is similar to the open-cycle evaporation process. The steam vaporizes the working fluid of a closed-cycle loop on the other side of an ammonia vaporizer. The vaporized fluid then drives a turbine that produces electricity. The steam condenses within the heat exchanger and provides desalinated water.



Hybrid cycle OTEC plant

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Factors to be considered while choosing a site:

- Thermal gradient in the ocean
- Topography of the ocean floor
- Seismic activity
- Availability of personnel to operate the plant
- Infrastructure – airports, harbors, etc.
- Local electricity and desalinated water demand.
- Political, ecological constraints
- Cost and availability of shoreline sites

Best sites are in the tropical belt between 20 degrees North and South

OTEC Application

- OTEC can be used to generate electricity, desalinate water, support deep-water Mari culture, and provide refrigeration and air-conditioning, mineral extraction. These

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complementary products make OTEC systems attractive to industry and island communities even if the price of oil remains low

- OTEC can also be used to produce methanol, ammonia, hydrogen, aluminum, chlorine, and other chemicals.
- Deep-Water-Supported Mariculture

  Deep-drawn seawater from an OTEC plant is cold, rich in nutrients, relatively free of pathogens, and available in large quantities. It is an excellent medium for growing phytoplankton and microalgae, which in turn support a variety of commercially valuable fish and shellfish. The large, constant flow of water pumped from an OTEC plant will reduce disease and contamination in the ponds; marine life, therefore, can be grown in high densities. In addition, deep-drawn cold water can be mixed with warm surface water, allowing local communities to culture a broad variety of species.

- Desalinated Water

  Desalinated water can be produced in open- or hybrid-cycle plants using surface condensers. In a surface condenser, the spent steam is condensed by indirect contact with the cold seawater. This condensate is relatively free of impurities and can be collected and sold to local communities where natural freshwater supplies for agriculture or drinking are limited.

- Refrigeration and Air-Conditioning

  The cold [5°C (41°F)] seawater made available by an OTEC system creates an opportunity to provide large amounts of cooling to operations that are related to or close to the plant. The cold seawater delivered to an OTEC plant can be used in chilled-water coils to provide air-conditioning for buildings.

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Benefits of OTEC

- No fuel burned, carbon di-oxide emission - less than 1% of fossil fuel plant: has significant potential to provide clean, cost-effective electricity for the future
- Supplies steady power independent of weather fluctuations or seasonal variations
- Machinery requires only only slight modifications to existing well tried ones ie heat exchanger, turbines etc
- Nutrient rich cold water promotes mariculture
- Produces desalinated water for industrial, agricultural, and residential uses; Fresh water production (1 MW plant -> 4500 m^3)
- Cold water for air conditioning or refrigeration

Disadvantages of OTEC(Environmental Impacts)

- An OTEC facility requires a substantial initial capital outlay
- OTEC has not been demonstrated at full scale over a prolonged period with integrated power, mariculture, fresh-water, and chill-water production.

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- OTEC is only feasible at relatively isolated sites (deep tropical oceans); from such sites, the power and marine products must be transported to market.
- OTEC is ecologically controversial--at least untested--in large scale and over a long period.
- The capital cost of installation is still extremely high and at the present time the technology remains at the planning/feasibility study stage.

OTEC Power Stations in the World

- Hawaii, USA: A 105 kW plant at the Natural Energy Laboratory of Hawaii, which is closed-cycle, provides power for about 120 homes.
- Saga University, Japan: This University operates a 50 kW double-Rankine system for demonstrations and research.
- Kavaratti, India: The National Institute of Ocean Technology is establishing a 65-kilowatt experimental plant.
- Qingdao, China: A 10 kW plant was developed in 2017.
- Other projects: There is ongoing exploration and development of OTEC in countries like Turkey, Jamaica, and the Turks and Caicos Islands.

Introduction to Geothermal Energy

Geothermal energy is the thermal energy present in the interior of the earth. Geothermal energy can be extracted from earth's interior in the form of heat. Volcanoes, geysers and hot springs are visible signs of the large amounts of heat lying in earth's interior. The geothermal energy from earth's interior is almost inexhaustible. Although the amount of thermal energy within the earth is very large, but useful geothermal energy can be extracted at only certain suitable sites.

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Geothermal Power Plants

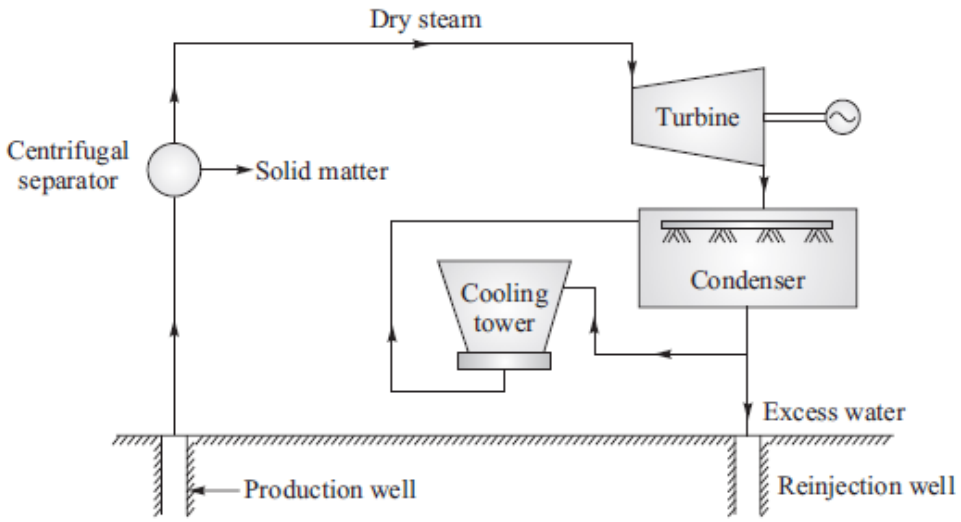
Hydrothermal resources are formed when underground water has access to high-temperature, porous rocks, capped by a layer of solid impervious rock. The entrapped water is heated by surrounding rocks. The hydrothermal resources can be classified depending on the heating of entrapped water as dry steam fields or vapour dominated resources, wet steam fields or liquid dominated resource and hot water resource. Vapour dominated fields can deliver steam with almost no water. Liquid dominated fields can deliver the mixture of steam and hot water, while hot water resource can only deliver hot water.

Vapour Dominated Plant

Water boils underground in a hydrothermal resource when it has pressure of about 7 atm and temperature of about l65°C. The plant consists of (i) production well to extract steam from the hydrothermal resource, (ii) a centrifuge separator to remove solid matter from the steam, (iii) a turbine to convert thermal energy into mechanical energy, (iv) a generator coupled to turbine to generate electric power, (v) a condenser to condense wet steam exited from turbine into water by direct contact with cooling water and (vi) a cooling tower to cool warm water

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exited from the condenser and returning the cooled water to the condenser. The excess water is disposed of in the reinjection well.. The cycle used in this system is called Rankine cycle.



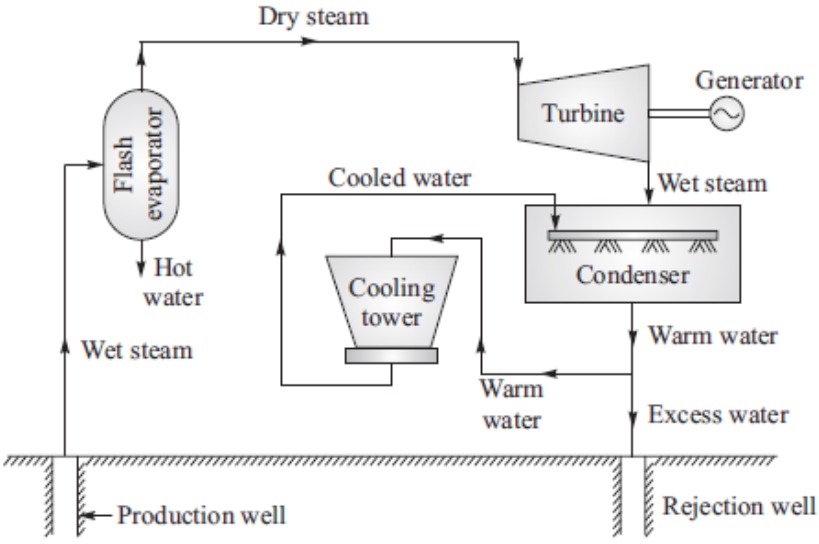
Dry steam hydrothermal plant.

Wet steam or liquid dominated

The wet steam or liquid dominated fields or resources can be further divided into high temperature (above l75°C) enabling the use of flash steam evaporator to produce dry steam and low temperature (range of 95–l75°C) where geothermal heat is used to vaporise a violatic refrigerant.

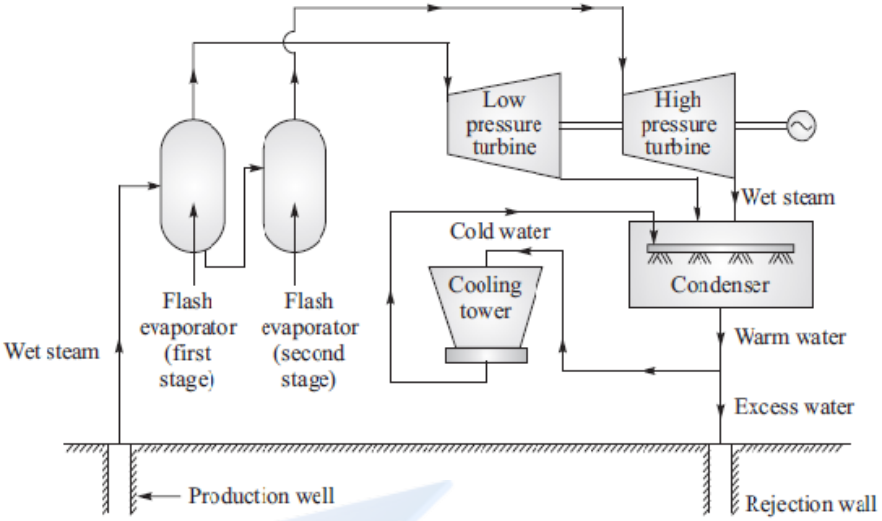
a) Liquid Dominated High-Temperature Plant or System.

The type of plant or system is used where hydrothermal reservoir has temperature and pressure of 230°C and 40 atm respectively. The plant consists of flash evaporator to obtain dry steam from high temperature wet steam by lowering pressure in it, turbine with directly coupled generator to extract energy from dry steam, condenser to condense used steam into water and cooling tower to cool the warm water. The excess water is disposed of in the ground .



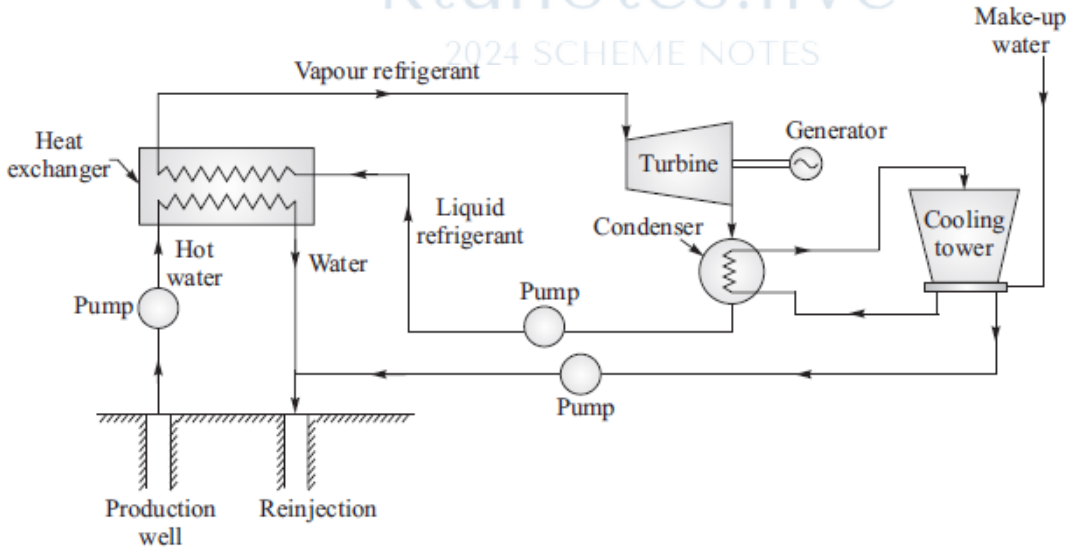
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In dual flash system, the hot water is flashed in two flash evaporators in two stages. Steam obtained in the first stage is used in low pressure turbine and steam is again flashed to obtain high pressure to be used in high pressure turbine .



b) ***Liquid dominated low-temperature plant or system.***

The water available from the hydrothermal reservoir is at low temperature range of 90–l75°C, which is insufficient to produce steam using a flash evaporator. A low boiling point refrigerant is used as a working medium. The plant runs on binary cycle. The refrigerant is evaporated in a heat exchanger using the heat of water obtained from hydrothermal reservoir. The refrigerant vapour runs the turbine with a generator coupled to it. The used refrigerant vapour is condensed in a condenser. The cooled water for cooling in condenser is obtained from a cooling tower.



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Advantages of Geothermal Energy

- It is a reliable source of energy.
- It is in continuous supply.
- Its availability is independent of weather.
- It has an inherent storage capability and it does not require any storage device.
- Small space is required to install the plant.
- It does not require any supply of fuel to generate heat.
- It is the least polluting.
- It is versatile in use.
- It is cheaper than other energy sources.

Disadvantages of Geothermal Energy

- Geothermal energy is available as low-grade heat, that is, temperature of geothermal fluid is low.
- Geothermal fluids also bring dissolved gases and solute, which lead to air and land pollution.
- Removal of heated water from the hydrothermal reservoir may lead to land subsidence or seismic imbalance.
- Drilling and blasting of wells may cause instability of land structure and may develop risk of any earthquake or seismic imbalance.
- Geothermal energy cannot be transported over long distances (less than 30 km).
- The life of plant equipment is limited due to corrosive and abrasive nature of geothermal fluids.

Criteria for Selection of Geothermal Site

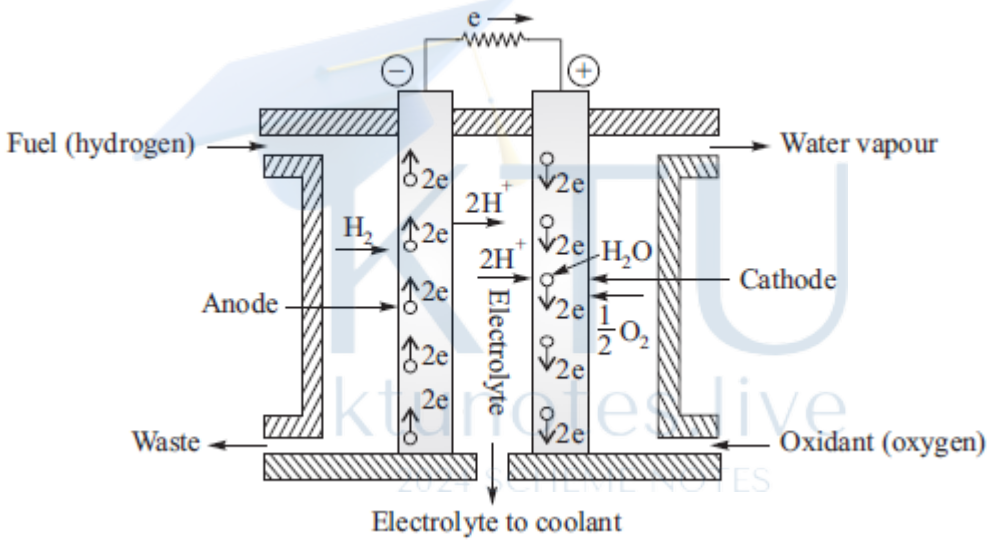
- Vapour dominated geothermal site is preferred as it is possible to obtain dry steam at about 200°C and pressure of 8 atm.
- Site should facilitate easy drilling to reach geothermal energy resource with least cost.
- Site should have sufficient geothermal energy to run a plant of adequate power capacity.
- Site should be near the load centre to reduce the cost of transmission.
- The dissolved gases and solids in the geothermal fluids should be as little as possible.

Fuel Cell

A fuel cell converts chemical energy of a fuel directly into electrical energy. Fuel gas diffuses through the anode and it gets oxidized. On oxidation, the fuel gas releases electrons to the external circuit. The oxidant gas diffuses through cathode and it is reduced by the electrons coming from external circuit to cathode. The fuel cell keeps the fuel and oxidant separate without mixing, but it allows the transfer of electrons from the fuel to the oxidant by an external electric circuit containing a load.

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The fuel cell consists of two electrochemically conducting electrodes separated by an electrolyte. The arrangement of fuel cell components is shown in Fig. The electrodes are made of porous nickel material to collect charges and concentrated phosphoric acid is filled between the electrodes, which acts as the electrolyte. The pores enable better contact between gas, electrolyte and electrode for faster electrochemical reaction. The fuel (hydrogen) is fed into the anode side of the cell. The fuel (hydrogen) is oxidised and liberated electrons move to the external circuit. The remaining positive hydrogen ions move from the anode into the electrolyte through porous cell walls. The oxidant (oxygen) is fed into the cathode side, where it is reduced by the electrons coming from anode through external circuit. The remaining negative oxygen ions enter the electrolyte from the porous wall of cathode. The negative oxygen ions and positive hydrogen ions combine to form water. The electrochemical reactions are as follow:



Anode

$$2H_2 \longrightarrow 4H^+ + 4e$$

Cathode

$$O_2 + 4e \longrightarrow 2O^{2-}$$

Electrolyte

$$4H^+ + 2O^{2-} \longrightarrow 2H_2O$$

Advantages of Fuel Cells

- It is quiet in operation as it is a static device,
- It is less pollutant,
- Its conversion efficiency is more due to direct single stage energy conversion,

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- Fuel cell plant can be installed near point of use; thus transmission and distribution losses are avoided,
- No cooling water is needed as required in condenser of a conventional steam plant. The heat generated can be easily removed and discharged to atmosphere or used locally,
- Because of modular nature, any voltage/current level can be realized and the capacity can be added later on as the demand grows,
- Fuel cell plants are compact and require less space,
- Availability of choice from large number of possible fuels,
- Can be used efficiently at part load from 50 per cent to 100 per cent
- No charging is required.

Applications of Fuel Cells

- Fuel cells can be effectively used for load leveling.
- A central station power plant using fuel cell is also possible using gasified coal as fuel.
- Fuel cells are also suited for dispersed generation. By locating the fuel cells near load center, transmission and distribution cost would be avoided/ reduced, although there would be some cost for transporting the hydrogen.
- To meet the demand of isolated sites such as construction sites, military camps and small village community or hamlet, fuel cells are more suited than diesel generator set.
- For remote and inaccessible locations fuel cell can be used unattended for a long period.
- Emergency/auxiliary supply to critical loads such as hospitals, etc., can be better met using fuel cells as compared to diesel generator set.
- Fuel cells can also be used as a mobile power source in vehicles, submarines, spacecraft, etc.
- Fuel cells are also proposed as a power source for propulsion of electric vehicles.
- Fuel cells can be used to power portable electronic devices as substitute for primary or re-chargeable batteries.

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Classification of Fuel Cells

- ***Based on the Type of Electrolyte***
  a) Phosphoric Acid Fuel Cell (PAFC)
  b) Alkaline Fuel Cell (AFC)
  c) Polymer Electrolytic Membrane Fuel Cell (PEMFC) or Solid Polymer Fuel Cell (SPFC) or Proton Exchange Membrane Fuel Cell (PEMFC)
  d) Molten Carbonate Fuel Cell (MCFC)
  e) Solid Oxide Fuel Cell (SOFC)

- ***Based on the Types of the Fuel and Oxidant***
  a) Hydrogen (pure) – Oxygen (pure) fuel cell
  b) Hydrogen rich gas – air fuel cell
  c) Hydrazine – Oxygen/hydrogen peroxide fuel cell
  d) Ammonia – air fuel cell
  e) Synthesis gas – air fuel cell
  f) Hydrocarbon (gas) – air fuel cell

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(g) Hydrocarbon (liquid) – air fuel cell

- ***Based on Operating Temperature***
  (a) Low temperature fuel cell (below 150 °C)
  (b) Medium temperature fuel cell (150 °C–250 °C)
  (c) High temperature fuel cell (250 °C–800 °C)
  (d) Very high temperature fuel cell (800 °C–1100 °C)

- ***Based on Application***
  (a) Fuel cell for space applications
  (b) Fuel cell for vehicle propulsion
  (c) Fuel cell for submarines
  (d) Fuel cell for defense applications
  (e) Fuel cell for commercial applications

- ***Based on the Chemical Nature of Electrolyte***
  (a) Acidic electrolyte type
  (b) Alkaline electrolyte type
  (c) Neutral electrolyte type

Alkaline Fuel Cell

- **Fuel:** hydrogen or hydrogen-rich gas
- **Oxidant:** oxygen or air
- **Electrodes:** porous nickel
- **Electrolyte:** KOH(40%)
- **Reactions**

$$\frac{1}{2}O_2 + H_2O + 2e \longrightarrow 2OH^- \text{ (cathode)}$$

$$H_2 + 2OH^- \longrightarrow 2H_2O + 2e \quad \text{(anode)}$$

It uses H_2 or H_2 rich gas as the fuel and oxygen or air as the oxidant. 40% aqueous KOH solution is used as the electrolyte. The operating temperature is about 90^0C. The hydrogen gas at anode is oxidised, resulting in the liberation of electrons. Electrons are forced through external circuit to cathode. At cathode, oxygen gas, water and electrons combine to produce OH^- ions. The reaction is given as follows:

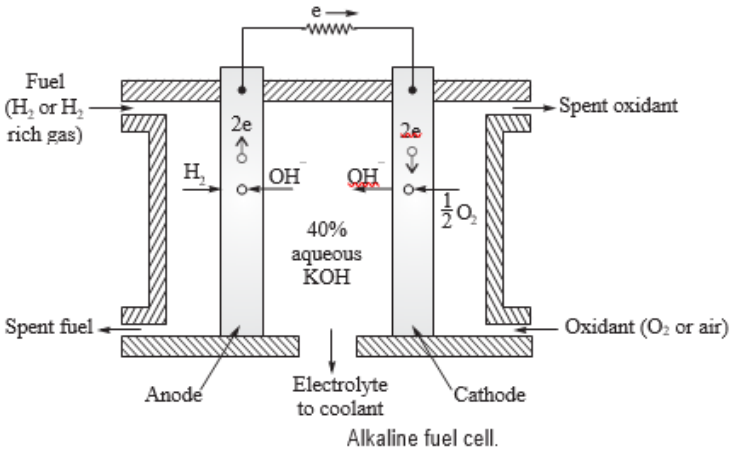
$$\frac{1}{2}O_2 + H_2O + 2e \longrightarrow 2OH^-$$

These OH^- ions move from cathode to anode through the electrolyte, where these combine with hydrogen gas to produce water. The reaction is as follows:

$$H_2 + 2OH^- \longrightarrow 2H_2O + 2e$$

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Alkaline fuel cell.

. It has porous electrodes of nickel separated by the electrolyte consisting of a solution of KOH (40%) which also helps in preventing the reactants (hydrogen and oxygen) from directly interacting with each other. The fuel used for the alkaline fuel cell has to be free from carbon dioxide because carbon dioxide can combine with electrolyte (potassium hydroxide) to form potassium carbonate. The potassium carbonate increases the resistance to motion of OH^- ions, thereby decreasing the output voltage of the cell.

If air is used as the oxidant in place of oxygen, the air must be free of carbon dioxide as the presence of carbon dioxide lowers the performance of cell.

Phosphoric Acid Fuel Cell(PAFC)

- **Fuel**:hydrogen
- **Oxidant**: oxygen
- **Electrolyte**: phosphoric acid
- **Electrodes**: porous nickel
- **Reactions:**

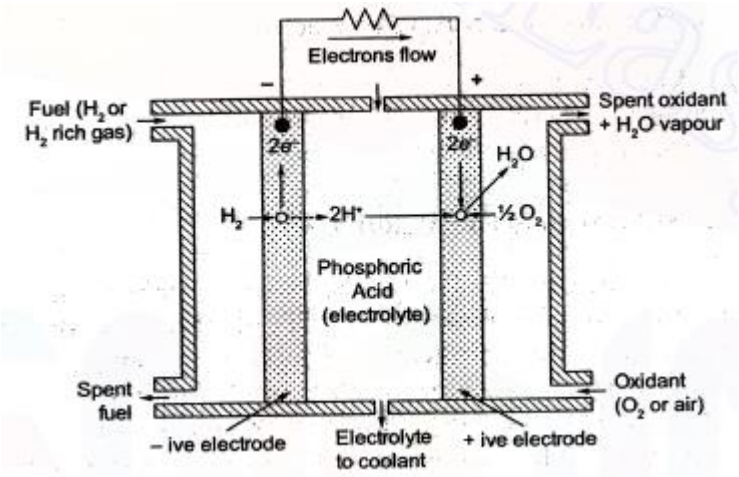
$$H_2 \longrightarrow 2H^+ + 2e^- \quad \text{(anode)}$$

$$2H^+ + \frac{1}{2}O_2 + 2e^- \longrightarrow H_2O \quad \text{(cathode)}$$

- **Output:** l.23 V at 25^0C
- **Efficiency:** 83%

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It consists of two electrodes of porous conducting material (commonly nickel) to collect charge, with concentrated phosphoric acid filled between them, to work as an electrolyte. Pure hydrogen or a hydrogen rich gas is supplied at the negative electrode and oxygen or air is supplied at the positive electrode. The pores provide an opportunity for gas, electrolyte and electrode to come into contact for electrochemical reaction. The reaction is normally very slow and a catalyst is required in the electrode to accelerate the reaction.

Platinum serves as the best catalyst for both electrodes and used for premium fuel cells. A less expensive material such as nickel for negative electrode and silver for positive electrode is used wherever possible. Thus finely divided platinum or nickel/silver deposited on the outer surface of electrodes are used as catalyst. during usage of cell the catalyst gradually loses its activity. This loss of activity is often attributed to poisoning of the catalyst by the impurities in the fuel.

At the negative electrode hydrogen gas is converted to hydrogen ions(H^+) and an equal number of electrons(e^-). Thus,

$$H_2 \longrightarrow 2H^+ + 2e^-$$

The electrons originating at the negative electrode flow through the external load to the positive electrode. Also the H^+ ions migrate from the negative electrode towards the positive electrode through the electrolyte. On reaching the positive electrode, they interact with O_2 to produce water. Thus,

$$\tfrac{1}{2} O_2 + 2H^+ + 2e^- \longrightarrow H_2O$$

Combining the above equations indicates that a fuel cell combines H2 and O2 to produce water and electrical energy.

The overall reaction is

$$H_2 + \tfrac{1}{2} O_2 \longrightarrow H_2O$$

The operating temperature of PAFC is 150^0C to 200^0C.at atmospheric pressure it produces an ideal e.m.f of 1.23 V at 25^0C.

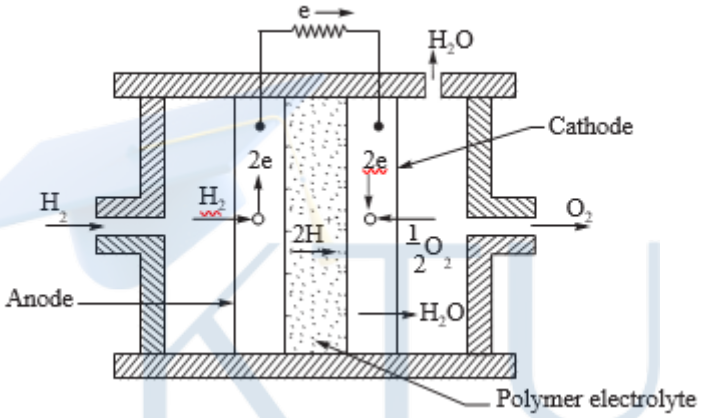
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Polymer Electrolyte Membrane Fuel Cell (PEMFC) or Proton Exchange Membrane Fuel Cell or Solid Polymer Fuel Cell

- **Fuel**: hydrogen
- **Oxidant**: oxygen
- **Electrodes**: deposited platinum layers
- **Electrolyte**: proton conducting polymer membrane
- **Reaction**

  $H_2 \longrightarrow 2H^+ + 2e^-$ (anode)

  $\frac{1}{2} O_2 + 2H^+ + 2e^- \longrightarrow H_2O$ (cathode)
- Output: l.23 V at 25°C.



The fuel cell uses a membrane of organic materials such as polystyrene sulphonic acid as electrolyte. The membrane has property to allow H^+ ions to pass through it. The desired properties of membrane are as follow:

- High ionic (H+) conductivity
- Non-permeable to reactant gases such as oxygen and hydrogen
- High resistance to dehydration
- Lesser tendency to electroosmosis
- High amount of mechanical stability

The fuel cell consists of a thin layer of electrolyte membrane which has platinum deposited on its each surface to act as electrodes (anode and cathode). H_2 enters and interact with anode. The gas is converted into H^+ ions and electrons are liberated.

The reaction is given by

$$H_2 \longrightarrow 2H^+ + 2e^-$$

Hydrogen ions formed at anode are transported to cathode through photon exchange membrane and electrons are forced through the outer circuit to cathode. At cathode, ions, electrons and oxygen gas interact to produce water

The reaction is

$$2H^- + 2e^- + 1\ O_2 \longrightarrow H_2O$$

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The cell produces emf of l.23 V at 25°C. It can operate at the temperature range of 40–60°C.

Molten Carbonate Fuel Cell

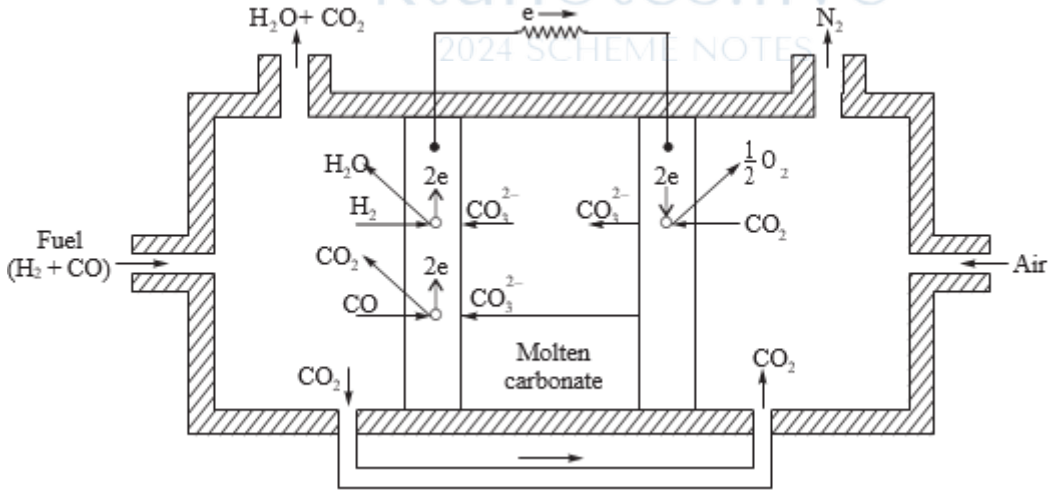
- **Fuel:** synthetic gas (H_2 + CO)
- **Oxidant:** air
- **Electrodes:** nickel (anode) and silver (cathode)
- **Electrolyte:** molten carbonate (sodium bicarbonate)
- **Reaction:**

$$H_2 + {CO_3}^{2-} \rightarrow H_2O + CO_2 + 2e^- \text{ (anode)}$$
$$CO + {CO_3}^{2-} \rightarrow 2CO_2 + 2e^- \text{(anode)}$$
$$O_2 + 2CO_2 + 4e^- \rightarrow 2{CO_3}^{2-} \text{ (cathode)}$$

- **Output**: l V at 700°C

The molten carbonate fuel cell has a high operating temperature with molten carbonate mixture as electrolyte. It offers the prospect of the use of a variety of fossil fuels including coal. The special feature of these cells is that they can oxidise carbon monoxide to carbon dioxide as well as hydrogen to water during their operation. Hence, the cell can use inexpensive mixture of hydrogen and carbon monoxide, which is called synthetic gas. Also, the presence of carbon dioxide in fuel and air does not have any adverse effect on the working of the cell.

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The carbonate of alkali metals (Na, K and Li) in molten state is used as the electrolyte.This necessity makes the cell to operate at a temperature above the melting point of carbonates (range of 600–700°C). The porous nickel and silver are used as the anode and the cathode electrode respectively which are separated by electrolyte held by a sponge-like ceramic matrix.

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The synthetic gas (H_2 + CO) is used as the fuel and air is used as the oxidant. The synthetic gas is passed through the anode, where the hydrogen and carbon monoxide are oxidised with CO_3^{2-} ions, thereby liberating electrons. The electrons move to cathode from the external circuit. At the cathode, oxygen gets reduced in the presence of carbon dioxide and electrons, thereby forming CO_3^{2-} ions.

The reactions are as follows:

$$H_2 + CO_3^{2-} \rightarrow H_2O + CO_2 + 2e^- \text{ (anode)}$$
$$CO + CO_3^{2-} \rightarrow 2CO_2 + 2e^- \text{(anode)}$$
$$O_2 + 2CO_2 + 4e^- \rightarrow 2CO_3^{2-} \text{ (cathode)}$$

The e.m.f generated by the cell is about l V at 700°C. Any fuel which can be converted into a mixture of hydrogen and carbon monoxide can be used. However, the mixture has to disulphurized as sulphur can poison the electrodes or reduce their effectiveness as a catalyst. The discharge from the reactants consists of steam, carbon dioxide and nitrogen and the discharge has temperature of 540°C. This hot discharge can be used to generate power usingwaste heat boiler with steam turbine to run generator.

Solid Oxide or Ceramics Fuel Cell (SOFC)

- **Fuel:** synthetic gas (H2 + CO)
- **Oxidant**: air
- **Electrodes:** porous platinum ceramic
- **Electrolyte:** ceramic (zirconium oxide) conducting oxygen ions

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- **Reaction:**

$$H_2 + O^{2-} \longrightarrow H_2O + 2e^- \quad \text{(anode)}$$
$$CO + O^{2-} \longrightarrow CO_2 + 2e^- \quad \text{(anode)}$$
$$O_2 + 4e \longrightarrow 2O^{2-} \quad \text{(cathode)}$$

- **Output:** l V at 800–l000°C

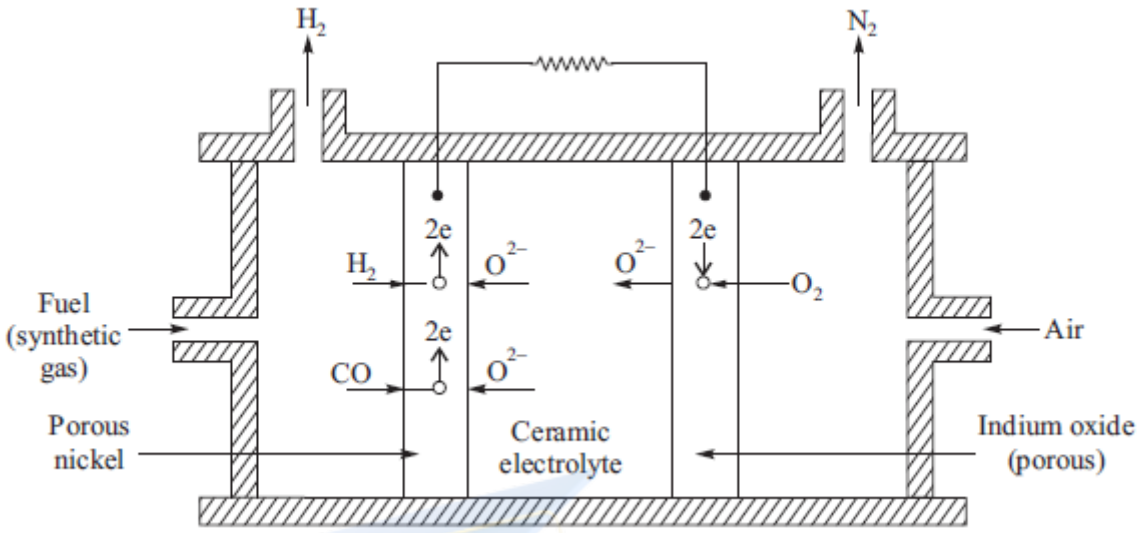
It has been found that certain solid oxides or ceramics can be used as an electrolyte and these ceramics can conduct oxygen ions at a high temperature. Zirconium oxide is one such ceramic. The fuel cell has porous nickel as anode electrode and indium oxide as cathode electrode.

The operating temperature of the cell ranges from 600 to l000°C. The fuel is a synthetic gas, that is, a mixture of hydrogen and carbon monoxide. At the anode, hydrogen and carbon monoxide react with oxygen ions present in the electrolyte to produce carbon dioxide and water. Hydrogen and carbon monoxide liberate electrons on oxidation. The liberated electrons flow through external circuit to cathode. At the cathode, oxygen is reduced by electrons to oxygen ions.

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The reactions are as follow:

$$H_2 + O^{2-} \longrightarrow H_2O + 2e^- \quad \text{(anode)}$$
$$CO + O^{2-} \longrightarrow CO_2 + 2e^- \quad \text{(anode)}$$
$$O_2 + 4e \longrightarrow 2O^{2-} \quad \text{(cathode)}$$



Regenerative Fuel Cell

Regenerative or reversible fuel cell is a cell in which the reactants (fuel and oxidant) are regenerated from the products formed from the oxidation and reduction of fuel and oxidant respectively. It implies that the reactants are regenerated from its products so that these can be recycled into the fuel cell. Regeneration is carried out within or external to the fuel cell.

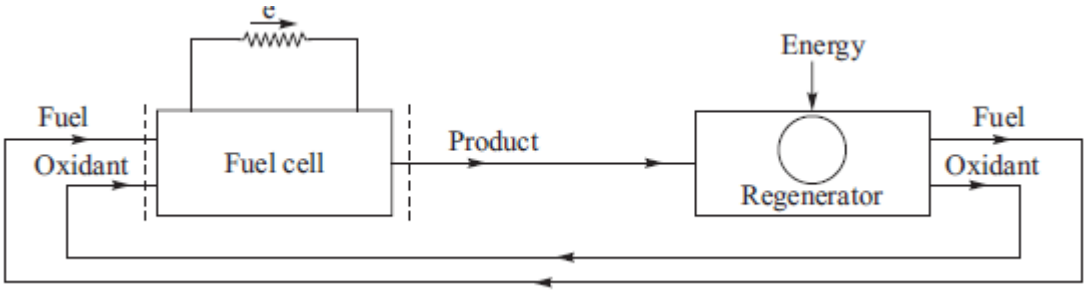
The regeneration can be carried out using chemical, electrical, thermal, radioactive and photochemical method.

In a regenerative fuel cell, reactants are converted into products with the removal of electrical work (W) and the products are converted into reactants in a regenerator at a higher temperature (T_H). Hence, products can be considered as working fluid and fuelcell with regenerator forms a heat engine cycle, that is, heat engine performing cycle to givework output using heat supplied.

The efficiency of regenerative fuel cell is, therefore, limited to Carnot engine efficiency. The efficiency of regenerative fuel cell is the product of efficiencies of a fuel cell and a regenerator. The efficiency of regenerative fuel cell is in the range of 5–20%.

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Limitations of Fuel Cell

- Capital cost of fuel cell is high.
- Heavy corrosion of electrodes causes low lifespan of a fuel cell.
- Degradation of electrodes and electrolyte reduces the performance of a fuel cell.

Hydrogen Energy

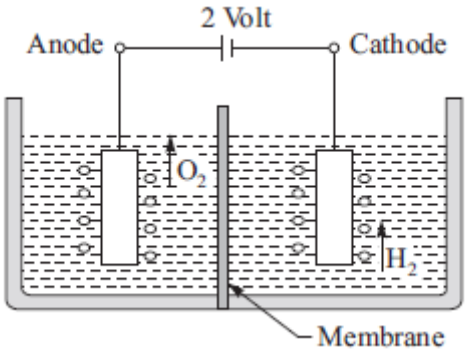
Hydrogen is considered an ideal fuel for future as it has the potential to be used as fuel for running cars and rockets, in meeting the energy needs of buildings and generating the electricity to meet the needs of industry. Free hydrogen does not occur naturally in nature. It has to be generated from some other energy sources. Hence, it cannot be classified as a primary source of energy. Hydrogen is always obtained using some primary source and it is, therefore a secondary source.

Production of Hydrogen Energy

The various methods of production of hydrogen include electrolysis of water, thermochemical or steam reforming of methane, thermal decomposition or thermolysis of water and biophotolysis.

Electrolysis of Water

It is the simplest method of hydrogen production. The method uses an electrolytic which consist of two electrodes immersed in an aqueous conducting solution called electrolyte. When a direct current is passed through the cell, it decomposes waterinto hydrogen and oxygen. Oxygen is formed at anode while hydrogen is formed at cathode. Metal or carbon plates are used as electrodes. The aqueous KOH solution is used as the electrolyte. A decomposition voltage of 2 V is applied.



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The chemical reactions of decomposition of water are as follows:

$$4\ H_2O + 4\,\bar{e} \rightarrow 2H_2 \uparrow + 4OH \text{ (cathode)}$$

$$4\ OH^- \rightarrow O_2 \uparrow + 2H_2O + 4\,\bar{e} \text{ (anode)}$$

Storage of Hydrogen

Hydrogen can be stored in the following forms:

- Hydrogen gaseous form
- Hydrogen liquid form
- Metallic hydrides form
- Chemical hydrides form

Hydrogen gaseous form of storage

As hydrogen has lower density compared to natural gas, it requires 3.6 times the volume occupied by natural gas to store the same mass of gas at the same pressure. The hydrogen can be stored in strong steel tanks or cylinders like natural gas, but storage pressure is kept high in the range of 350–750 atm so that sufficient mass of gas can be stored. This method is suitable where a small amount of hydrogen is required. Large amount of hydrogen is generally stored in the underground facilities.

Hydrogen liquid form of storage

Hydrogen can be stored in the liquid form as liquid takes less storage space. However, storage space in liquid form requires cryogenic or temperature lower than 20 K (–253°C). This low temperature has to be maintained for the complete storage duration, which makes this type of storage a costly preposition.

Metallic hydrides form of storage

Hydrogen chemically joins with certain metals and forms hydrides. When hydrides are heated, the hydrogen gas can be liberated. The pressure of gas released depends on temperature and it remains constant at a certain temperature. The metallic alloys used for forming hydrides are lanthanum nickel, iron titanium and magnesium nickel.

$$La\ Ni_5 + 3H_2 \rightleftharpoons (La\ Ni_5)\ H_6 + \text{heat}$$

$$Fe\ Ti + H_2 \rightleftharpoons (Fe\ Ti)\ H_2 + \text{heat}$$

$$Mg_2\ Ne + 2H_2 \rightleftharpoons (Mg_2\ Ni)\ H_4 + \text{heat}$$

Owing to the heavy weight of all metallic alloys, hydrides storage is unsuitable to be used for running of vehicles. Metallic hydride storage is desirable when metal alloys forming hydrides is

- Inexpensive,
- Hydrides are able to contain a large amount of hydrogen,
- Storage of hydrogen in metallic alloy can be done at the room temperature
- Significant pressure of liberated gas is possible at the room temperature.

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Chemical hydride form of storage

In chemical hydride method, chemical compounds are used to form hydrides in alkaline solution. Chemical hydride process is a safer method compared to metallic hydride from of storage. It is also a more volumetrically efficient storage method.

Merits and Demerits Of Hydrogen as a Fuel.

Merits

- Exists in an unlimited supply
- Renewable
- Is clean- burning (water is the by product and no CO2 is released)
- Weighs less and generates more power than hydrocarbon based fuels.
- Burns faster and at lower temperature as compared to gasoline.

Demerits (problems associated with hydrogen energy)

- Difficult to contain as tanks are bulky
- Reacts with metal in tanks causing them to become brittle
- Is explosive
- Establishing distribution facility will cost trillions of dollars.

Applications of Hydrogen

- Hydrogen is used as fuel in various types of fuel cells to generate electric power.
- Hydrogen is used as aviation fuel as it has high energy density.
- Hydrogen is used in catalytic combustion, thereby replacing LPG for domestic cooking.
- Hydrogen can be used as fuel in running vehicles as it has lesser pollutants on combustion.
- Hydrogen can be used as steam generator as it gives pure water on combustion.
- Hydrogen is used for cooling of large electrical generators and motors.
- Hydrogen is used for hydrogenation of vegetable oils.
- Hydrogen is used for manufacturing ammonia.
- Hydrogen is also used for manufacturing synthetic methanol and urea (ammonium nitrate).
- Hydrogen is used for manufacturing tungsten filaments of lamps.
- Hydrogen has a great potential to be used as intermediate means either to store energy or to transport energy to long distances.

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Previous University Questions

1. With a neat diagram explain hybrid cycle OTEC power plant. Also, list any two advantages of OTEC system.
2. With a neat diagram explain closed cycle OTEC power plant
3. List the advantages and limitations of tidal power plants.
4. Explain with the help of block diagram, the working of open cycle OTEC plants
5. Advantages and limitations of OTEC.
6. Explain with the help of block diagram, the working of closed cycle OTEC plants.
7. Enlist the factors to be considered for site selection of OTEC plants.
8. With a neat diagram explain closed cycle OTEC power plant.
9. List any 4 advantages and 4 disadvantages of OTEC systems.
10. Explain in detail the working of a closed cycle OTEC plant with relevant diagrams.
11. List out the site selection factors for a OTEC plant.
12. A hybrid OTEC system is more advantageous compared to closed or open-cycle systems. Justify the statement.
13. Contrast the phenomena and principles utilised for tidal energy conversion and OTEC.
14. Compare the construction, working, and advantages of Anderson and Claude cycle OTEC systems.
15. What are the criteria used in the selection of suitable sites for OTEC power plants and discuss the importance of each criterion.
16. Differentiate the working principle of a tidal energy conversion system and an ocean thermal energy conversion system. State the advantages and disadvantages of these schemes.
17. What makes a hybrid OTEC system preferable to the Anderson and Claude cycle-type OTEC? Explain its works with a block diagram.
18. A separate working fluid such as ammonia, propane, or Freon is used in addition to water for closed-cycle OTEC. With a neat diagram explain the working of the closed cycle OTEC highlighting the advantage of using the additional working fluid.
19. A hybrid cycle combines the features of both the closed-cycle and open-cycle systems. Justify.
20. List any 3 advantages and 3 disadvantages of OTEC systems

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